



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION

1ST SEMESTER 2024/2025

Instructions: Answer Question One and any other TWO Questions.

QUESTION ONE

- (a) Explain the distinction between the following terms:
- Measurement and evaluation.
 - Histogram and frequency polygon.
 - Percentile point and percentile rank.
 - Skewness and Kurtosis.
- (b) Indicate the suitable scales (or levels of measurement) for each of the following variables:
- The heights of all the KU students.
 - The total number of lecturers at KU.
 - The positions of learners in an end-semester examination.
 - The scores of Form 4 students in a Mathematics test.

QUESTION 2

Given below is a set of grouped data showing the performance of students on a Form 2 Biology test.

20-24	25-29	30-34	35-39	40-44	45-49	50-54
5	8	15	18	20	10	4

Using the above tabular information, compute the following measures of:

- (a) Central tendency:
- Mode
 - Median
 - Mean
- (b) Variability:
- Range
 - Variance
 - Standard deviation
- (c) Using the measures of central tendency obtained in (a) above, describe:
- Fully the distribution of scores
 - The performance of students on the test
- (d) Distinguish between the following types of scores:
- Z-score and standard score
 - T-score and stanine
 - Standardized scores and normalized scores

QUESTION 3

- (a) Differentiate between the following terms:
- Formative and summative evaluation
 - Norm-referenced and criterion-referenced testing
 - Face validity and content validity
 - Paper and pencil (written) and performance tests
 - Test-retest and split-half methods of estimating reliability
- (b) Explain the effect of each of the following on the quality of a test:
- Use of test specifications
 - Small or large number of items in the test
 - Mixing difficult, moderate and easy items in an appropriate proportion when preparing a test
- (c) Explain two methods that a teacher can use to reduce subjectivity in marking an essay test.

QUESTION 4

(a) State four factors that might affect the selection of an item format (or question type) that a teacher would like to use in his/her test.

(b)

- Define a report form.
- Briefly discuss as to how important the report form is to the following:
 - Student
 - Teacher
 - Parents
 - Administrator

SECTION B (30 MARKS)

QUESTION 5

(a)

- Define correlation.
- How is a scatter plot (or scatter diagram) useful in determining the relationship between the two variables X and Y? Explain.

(b) Given below are scores that were obtained by a group of ten Form 3 students on Mathematics and Chemistry tests:

Student	A	B	C	D	E	F	G	H	I	J
Mathematics	8	10	6	2	3	4	6	8	9	4
Chemistry	3	4	10	8	2	9	6	7	6	5

- Calculate the Pearson product moment correlation coefficient r_{xy} between Mathematics and Chemistry.
- Interpret your calculated value (r_{xy}).
- Give two assumptions underlying the Pearson product moment correlation coefficient, r_{xy}
- Compute the Spearman Rank Order correlation coefficient r_s for the above given data.
- State two assumptions for the spearman Rank order correlation coefficient, r_s .

QUESTION 6**(a)**

- i.** What is item analysis?
- ii.** Distinguish between item difficulty index (P) and item discrimination index (D).
- iii.** What properties are desirable for Item difficulty index and item discrimination index?

(b) The table below summarizes student's responses on two multiple choice items (or questions).

Item No.	Group	Options				Omits	Total
		A*	B	C	D		
1	Upper group	50	7	0	3		60
	Lower group	30	3	0	22	5	60
	Group	A	B	C*	D	Omits	Total
2	Upper group	9	4	43	4	0	60
	Lower group	28	0	32	0	0	60

For Item 1, A* is the Key

For Item 2, C* is the Key



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QUESTION ONE

- a)** Discuss any THREE rationales for trainee teachers to learn Educational Statistics & Evaluation. (6 marks)
- b)** Using relevant Classroom examples, distinguish between the following terms:
 - i.** Formative and Summative evaluation (4 marks)
 - ii.** Reliability and Validity (4 marks)
 - iii.** Norm-referenced Grading and Criterion-referenced Grading (4 marks)
 - iv.** Continuous and discrete variables (4 marks)
- c)** Using educational relevant examples, discuss the four levels of measurement (8 marks)

QUESTION TWO

Consider the following distribution of scores obtained from a Form 2 Physics test

Score	Frequency
50	2
40	4
30	6
20	4
10	3
0	1

- a)** Explain the concepts of mean, median and mode
- b)** Plot a bar graph for this distribution
- c)** What's the mean, mode and median for the distribution?
- d)** Comment on the shape of the distribution

QUESTION THREE

The following scores in mathematics and geography test were obtained by 12 form 2 students.

Subject	A	B	C	D	E	F	G	H	I	J	K	L
Mathematics	73	71	70	67	64	61	61	60	58	56	53	50
Geography	40	35	33	27	29	31	30	26	28	34	25	22

- a) Using educationally relevant examples, discuss TWO factors that can distort the size of a correlation coefficient **(6 marks)**
- b) Using educationally relevant examples, explain why we should be cautious in interpreting a large size of a correlation coefficient **(6 marks)**
- c) Using data in the above table, calculate the Pearson Product Moment Correlation Coefficient (r_{xy}) between Mathematics and Geography tests **(6 marks)**
- d) Interpret the value of r_{xy} **(2 marks)**

QUESTION FOUR

A test developer carried out a try-out of a multiple choice test on his students, and presented the data on the item in the table below:

GROUPS	ALTERNATIVES				
	A	B	C*	D	TOTAL
UPPER	4	2	21	3	30
LOWER	10	3	12	5	30